SPE-226846-MS

Evaluating the CO_2 Storage Potential of Scoriaceous Basalts in Saudi Arabia: Implications for the Saudi Green Initiative

Mobeen Murtaza, Center for Integrative Petroleum Research, King Fahd University of Petroleum & Minerals, Saudi Arabia; Monica Campos, Department of Geosciences, King Fahd University of Petroleum & Minerals, Saudi Arabia; Hasan J. Khan, Department of Petroleum Engineering, King Fahd University of Petroleum & Minerals, Saudi Arabia; Scott Whattam, Department of Geosciences, King Fahd University of Petroleum & Minerals, Saudi Arabia; Manzar Fawad and Muhammad Shahzad Kamal, Center for Integrative Petroleum Research, King Fahd University of Petroleum & Minerals, Saudi Arabia; Nabil Ali Saraih and Israa S. Abu-Mahfouz, Department of Geosciences, King Fahd University of Petroleum & Minerals, Saudi Arabia

 DOI 10.2118/226846-MS

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

While CO_2 mineralization is a crucial area of research for climate change mitigation, the potential of Saudi Arabian scoriaceous basalts has remained largely unexplored. In this study, we aim to assess these unique basalts as a potential CO_2 sink by conducting a longer aging study and, paving the way for Saudi Arabia and regions with similar geological formations to actively contribute to carbon sequestration and emission reduction, supporting sustainable development goals (SDGs) on Climate Action.

Scoriaceous basalts were sourced from the Harrat Uwayrid volcanic field in western Saudi Arabia. To simulate CO_2 mineralization conditions, these samples were aged in a brine-CO_2 environment at 50°C and 1450 psi over three months. Advanced imaging techniques, including micro-CT and scanning electron microscopy (SEM), were employed to evaluate the mineralization potential and structural alterations within the basalt. Pre- and post-aging assessments provided detailed insights into the mineralization process occurring within the basalt groundmass, enabling a comprehensive understanding of the changes induced by CO_2 exposure.

The Micro-CT analysis showed a significant increase in porosity for the scoriaceous basalt sample after three months in a brine-CO_2 environment, with porosity rising from an initial 5.72% to 14.89%. Furthermore, it showed that precipitation occurred along the sites of dissolution. SEM imaging showed clear signs of etching and dissolution, indicating the effective reactivity of these basalts with CO_2 under the aging conditions. These observations underscore that both dissolution and precipitation processes occurred, with ion release from basalt leading to stable carbonate formation and effective CO_2 trapping through mineralization.

This investigation is a part of the study that was first to assess the scoriaceous basalts of Saudi Arabia for CO_2 storage, highlighting their distinct potential for CO_2 reduction. It also offers a foundational roadmap for future CO_2 sequestration projects, supporting the objectives of the Saudi Green Initiative.

Keywords: Basalts, CO_2 Storage, CO_2 Mineralization, Saudi Green Initiate, Sustainable Development Goals

Introduction

The ongoing rise in average global atmospheric temperatures is a primary driver of climate change, significantly altering Earth's climate and weather patterns. This rapid transformation is largely attributed to persistent anthropogenic greenhouse gas (GHG) emissions, particularly carbon dioxide (CO_2), which traps heat within Earth's atmosphere. Among anthropogenic GHGs, CO_2 is particularly problematic due to its high atmospheric concentration and extended residence time (Ahmed Ali et al., 2020; Nunes, 2023).

In 2022, global CO_2 emissions from energy combustion and industrial processes reached a record 36.8 gigatonnes, according to the International Energy Agency(IEA, 2023). The combustion of fossil fuels directly contributes to rising global temperatures, significantly disrupting ecological and environmental systems worldwide(Redlin & Gries, 2021). Consequently, mitigating these emissions through targeted reductions in CO_2 output is imperative.

Carbon dioxide (CO_2) sequestration stands as a pivotal strategy in mitigating climate change by capturing and securely storing CO_2 to reduce its atmospheric concentration. This approach encompasses both natural methods, such as enhancing carbon storage in soils and forests, and engineered technologies like direct air capture, geological storage, and mineral carbonation. By preventing CO_2 from entering or remaining in the atmosphere, sequestration plays a critical role in reducing greenhouse gas emissions and achieving long-term climate goals. Among these, geological storage is particularly important due to its ability to permanently isolate large volumes of CO_2 in deep underground formations(Boruah, 2025). Potential storage sites encompass saline aquifers, depleted oil and gas reservoirs, coal seams, and igneous rock formations, each offering distinct advantages for carbon storage. This method is technologically mature, highly secure when properly managed, and capable of storing CO_2 for thousands of years without significant risk of leakage.

While various geological formations have been investigated for CO_2 sequestration, each presents distinct advantages and limitations. Coal seams and unconventional gas reservoirs offer favorable geochemical environments for CO_2 trapping, particularly through adsorption mechanisms. However, their limited storage capacities make them less viable for large-scale, long-term sequestration. Sedimentary formations, especially those dominated by minerals like quartz, also pose challenges due to slow reaction rates with CO_2, which hinder efficient mineral trapping. Deep saline aquifers, on the other hand, provide vast storage potential and are among the most widely considered options due to their widespread availability and favorable porosity and permeability. However, their long-term storage security depends on caprock integrity and ongoing monitoring, as they rely primarily on physical and solubility trapping mechanisms. In contrast, igneous rocks—particularly basaltic and ultramafic formations—are increasingly recognized for their exceptional suitability in mineralization-based CO_2 sequestration. These rocks are rich in reactive divalent metal cations such as magnesium and calcium, which facilitate permanent and stable carbonate formation upon reacting with CO_2(Murtaza et al., 2024). Although site accessibility and injectivity can be more challenging, the high reactivity and permanence offered by igneous rocks make them among the most promising geological media for durable and secure carbon storage. Several CO2 injection projects have been summarized by (Ye et al., 2025). The CarbFix project in Hellisheiði, Iceland, targets basalt formations at depths between 400–800 m, utilizing dissolved CO2, and has injected approximately 101,000 tonnes of CO2 since 2012 with a notable 95% conversion rate. The Wallula project in Washington, USA, involved supercritical CO2 (scCO2) injected into flood basalt at depths of 800–900 m, injecting 1000 tonnes between July and August 2013 and achieving a 60% conversion rate before ceasing operations in 2015. The Nagaoka project in Niigata, Japan, initially conducted for geological storage rather than in-situ mineralization, injected 10,405 tonnes of scCO2 into a sandstone aquifer at a depth of 1100 m from 2003 to 2005, now serving solely for monitoring. The CarbonX project in Guangdong, China, set to start in 2024, will inject dissolved CO2 into basalt reservoirs at depths of 300–500 m, aiming for annual injections over 1000 tonnes, although conversion rates are not yet determined.

Globally, carbon mineralization in ultramafic and mafic rocks could theoretically sequester up to 60 million Gt of CO_2 if the resources become economically accessible and fully carbonated (Kelemen et al., 2019). Ultramafic rocks, containing over 85% mafic minerals, possess substantial storage potential due to their high magnesium content and rapid dissolution rates, particularly of olivine compared to pyroxene(Raza et al., 2022). However, their limited geographic distribution and potentially low fracture permeability present challenges. Basalts, conversely, are more widely available, covering roughly 5% of Earth's terrestrial surface and much of the ocean floor. Their mineralogical composition and reactivity with water, primarily driven by pyroxene minerals, facilitate rapid mineralization processes, ensuring long-term stability and minimal leakage risks(Olajire, 2013).

Saudi Arabia is making significant advancements in carbon capture, storage, and sequestration (CCS) to achieve its sustainability goals under the Saudi Green Initiative (SGI), aiming to reduce carbon emissions by 278 million tons per annum (mtpa) by 2030. Saudi Aramco spearheads these efforts, notably through projects such as the Hawiyah facility, which processes approximately 45 million cubic feet of CO_2 daily, injecting it into the Uthmaniyah reservoir for enhanced oil recovery and long-term sequestration. Additionally, Saudi Aramco explores CO_2 sequestration via in-situ mineralization in volcanic rocks in Jazan.

The Harrat Uwayrid region (HARU), an extension of Harrat ar Rahah in Western Saudi Arabia, provides an optimal setting for basaltic sequestration studies(Whattam et al., 2025). Covering approximately 7,200 km² and extending 225 km from Tabuk to Al Ula, HARU uniquely overlays Paleozoic sandstones instead of the Arabian-Nubian Shield. It features about 25 exposed basalt flows, facilitating detailed geological research.

In this study, we investigate CO_2 mineralization in Saudi basalt over extended experimental durations of up to three months. This prolonged exposure allows for a comprehensive evaluation of the long-term physical, chemical, and mechanical alterations in basalt samples subjected to CO_2-brine environments.

This research specifically investigates basalt type from HARU: scoriaceous basalt (SB), employing advanced analytical techniques including micro-CT, SEM-EDX, and XRD. Findings from these analyses are expected to optimize strategies for secure and effective carbon storage in igneous rocks, addressing critical knowledge gaps regarding their long-term behavior under CO_2 exposure.

Methodology

The study characterizes and evaluates scoriaceous basalt (SB) from Harrat ar Rahah-Uwayrid, Saudi Arabia. Basalt is primarily composed of orthopyroxene (55.3%), clinopyroxene (40.3%), and olivine (25.7%), with minor amounts of dickite (2.5%) and magnesium calcite (0.1%). Basalt specimen was prepared as 20 × 20 × 3 mm chip, cleaned with deionized water, dried at 60°C, and purged with nitrogen gas to remove contaminants.

The SB samples were aged in a high-pressure reactor simulating subsurface conditions, maintained at 50°C and pressurized with CO_2 at 1450 psi for three months. The reactor was filled to 70% capacity with brine, the composition of which included $NaHCO_3$, Na_2SO_4, NaCl, $CaCl_2 \cdot 2H_2O$, and $MgCl_2 \cdot 6H_2O$ with TDS of 57000 ppm.

Post-aging, SB chips were rinsed, dried, and analyzed for mineralogical and geochemical changes. Microstructural changes were examined via X-ray micro-computed tomography (μCT) using an Xradia

Versa XRM-500. Initial scanning resolution for SB was 9.93 μm (140 kV, 0.0643 mA, 1.2 sec exposure, 2201 projections), with a post-reaction scan at 11.62 μm to capture finer alterations. μCT images were processed using Fiji-ImageJ software. Images were registered with BIGWARP, aligned through linear transformations, filtered, segmented (Huang thresholding), and analyzed for porosity.

Scanning electron microscopy (SEM) coupled with energy-dispersive X-ray spectroscopy (SEM-EDX) complemented μCT by detailing microscale surface changes, dissolution patterns, and secondary carbonate formation. SEM images were analyzed in ImageJ, utilizing Otsu's method for porosity quantification. These analyses confirmed mineralogical and structural transformations resulting from CO_2 exposure.

Results and Discussion

In SB, mineralogical changes over time highlighted significant shifts driven by the rock's higher porosity and reactive surface area. Olivine gradually decreased from an initial 25.7% to 22.2%, while clinopyroxene exhibited a steady increase from 40.3% to 44.4%. Orthopyroxene, notably reactive within the porous SB matrix, became undetectable after three months.

The emergence of secondary mineral phases marked important stages of mineral transformation. Anorthite appeared at 15.6% after three months, clearly indicating secondary crystallization processes. Concurrently, magnesium calcite progressively increased from 0.1% to 1.7%, reflecting sustained carbonate precipitation over the studied period. Nepheline, absent initially, emerged notably at 16.2% after three months, signifying the evolving geochemical environment and its influence on mineral formation. Additionally, dickite, a clay mineral, maintained a minor but consistent presence, slightly decreasing from 2.5% to 2.1%.

These observations clearly illustrate the mineralogical evolution within the SB rock, emphasizing a distinct mineralization pathway characterized by rapid reactivity of orthopyroxene, sustained carbonate precipitation, and significant secondary phase formation, driven by enhanced porosity and reactive surface area.

The pre-reaction μCT scan of SB sample (Figure 1) reveals a highly porous structure, characterized by pitting on the external rock surface and numerous internal openings, indicative of a scoriaceous basalt fabric. There are multiple minerals, abundant pore spaces, and multimodal gray value distribution. Variations in gray value intensities correspond to different mineral phases, which can be identified as fresh olivine (mid-gray, Mineral 1), iddingsite-altered olivine (brighter-gray, Mineral 2), and high-density Fe-rich oxides (high-gray region, bright spots). The bright spots correspond to likely magnetite or magnesioferrite, based on their high X-ray attenuation in Figure 1.

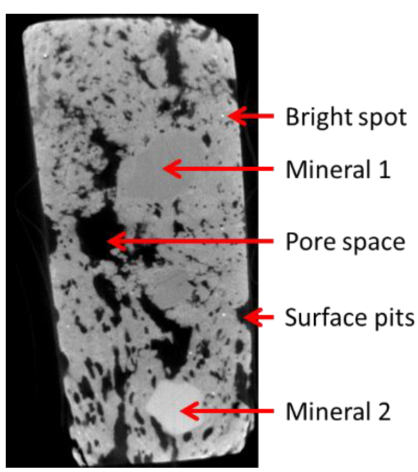

Figure 1—Pre-reaction SB μCT scans

The pore spaces and surface pits observed in the μCT scan are directly related to the scoriaceous texture of this basalt, reflecting its vesicle-rich nature and the presence of irregular cavities formed during magma degassing. These features appear as low-gray to black regions in the μCT scan, indicating areas of minimal X-ray attenuation due to their low-density composition and open porosity. These features contribute to enhanced permeability, allowing for increased fluid interaction during CO_2 exposure. The low-gray vesicles and black pore spaces suggest potential zones for carbonate precipitation. The initial porosity of the rock sample, which ignores the pitting structures found on the rock surface, is calculated as 5.7%.

Figure 2 presents the aligned pre- and post-reaction μCT scans of the SB sample, demonstrating a marked increase in porosity after three months of exposure to a CO_2-saturated brine environment. A comparative assessment of the scans reveals widespread mineral dissolution and noticeable expansion of the pore network, indicative of an increase in both reactive surface area and fluid accessibility.

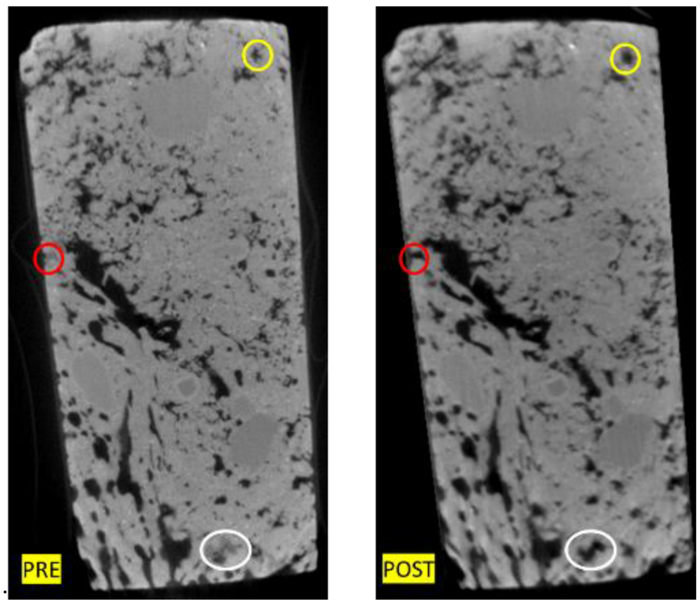

Figure 2—Comparison between pre- and post-reaction SB scans

A red circle highlights the dissolution of a surface mineral, most likely olivine, initially visible as a mid-gray phase in the pre-reaction scan. Its removal has led to enhanced surface roughness and improved permeability. Within the rock matrix, a white ellipse marks regions where olivine phenocrysts appear to have undergone extensive alteration or complete dissolution, resulting in low-gray voids that signal deeper fluid ingress and sustained leaching processes.

Additionally, the yellow circle draws attention to the enlargement of a vuggy structure located deeper within the sample, away from the immediate surface. This feature exemplifies matrix dissolution at depth and underscores the capacity of the reactive brine to penetrate and propagate dissolution fronts internally. The post-reaction scan reveals a porosity of 14.9%, reflecting a 1.6-fold increase compared to the initial state—evidence of substantial textural modification over the aging period.

The SEM images of the scoriaceous basalt (SB) illustrate its initial microstructure and the geochemical transformations induced after three months of treatment. Figure 3A (pre-aging) reveals a heterogeneous texture characterized by pre-existing voids that vary in morphology, including circular vesicles formed due to magma degassing, interconnected irregular voids within the matrix, and amygdales—vesicles that were subsequently filled with secondary minerals such as carbonates, zeolites, or clays during post-eruptive processes. Additionally, olivine phenocrysts, some displaying internal fractures, are embedded within the matrix. After three months of aging, Figure 3B (post-aging) exhibits notable dissolution features primarily affecting the olivine phenocrysts, where their internal fractures act as fluid pathways, accelerating the dissolution process. Figure 3C (composite image) provides a quantitative assessment of porosity evolution, with red regions representing the initial void space and green zones indicating areas of dissolution and secondary mineral precipitation. The distribution of newly altered zones highlights preferential weathering along crystal edges, inner fractures, and vesicle boundaries, demonstrating how geochemical alteration is facilitated by the rock's inherent microstructural heterogeneities. Porosity calculations using the Otsu-threshold method in Fiji/ImageJ indicate an increase from an initial 5.88% to 18.18% post-aging, aligning well with micro-CT-derived values.

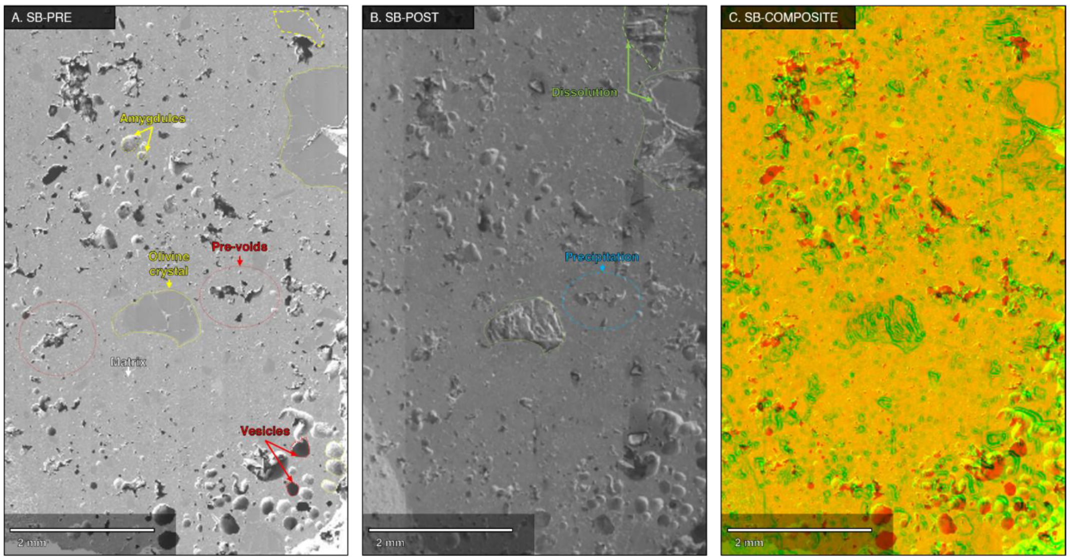

Figure 3—SEM images of SB pre- (A) and post-aging (B) and composite image (C).

Figure 4 provides evidence of secondary carbonate precipitation along pre-existing voids in the sample. Figures 4A and 4D illustrate an olivine phenocryst before and after the aging process, respectively. Notably, dissolution occurs preferentially along pre-existing fractures, accompanied by the formation of secondary mineral phases, likely carbonates, within these zones.

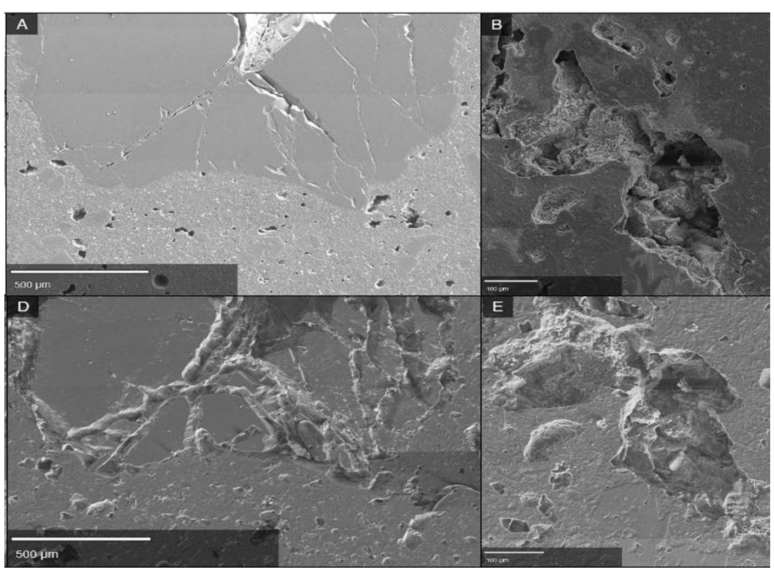

Figure 4—(A–B) presents SEM images of SB pre-treatment, while (D–E) illustrate post-treatment modifications, revealing mineralogical and textural changes.

In Figures 4B and 4E, an initially empty, irregularly shaped void within the basaltic matrix is depicted. Following exposure to the CO_2-rich brine, these void exhibits clear signs of secondary precipitation, characterized by alterations in surface topography and partial infilling. These features suggest localized saturation with respect to carbonate phases and highlight the dynamic interplay between mineral dissolution and precipitation during fluid-rock interaction.

EDS analysis of the SB after three months of CO_2 treatment reveals significant geochemical alteration indicative of mineral dissolution and secondary precipitation (Figure 5). The elemental mapping of a post-treatment olivine crystal highlights the spatial distribution of key elements, with magnesium (Mg) and silicon (Si) being prominently associated with olivine-rich regions, while iron (Fe) and calcium (Ca) indicate the presence of a secondary phase. The EDS spectrum provides a quantitative assessment of elemental composition, with oxygen (O, 34.8%), silicon (Si, 21.1%), sodium (Na, 9.4%), aluminum (Al, 7.6%), iron (Fe, 7.6%), platinum (Pt, 6.3%), calcium (Ca, 4.7%), titanium (Ti, 1.0%), nickel (Ni, 0.7%), and chlorine (Cl, 0.7%) being the dominant elements. The high magnesium (Mg) and silicon (Si) content is primarily attributed to the presence of olivine in the scoriaceous basalt. The presence of platinum is likely due to contamination from experimental instrumentation rather than a natural component of the basalt. Furthermore, aluminum and silicon content align with the presence of aluminosilicate phases, potentially forming amorphous silica or clay minerals.

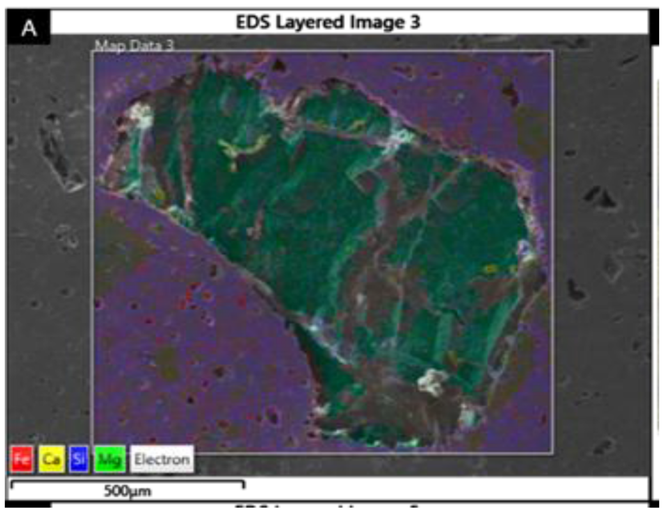

Figure 5—EDS analysis of SB post-treatment after three months of CO_2 exposure, illustrating mineral dissolution and secondary precipitation.

Conclusions

In this study, the CO_2 mineralization potential of scoriaceous basalt (SB) was evaluated following three months of exposure to a CO_2-rich brine environment, utilizing analytical techniques including SEM and micro-CT. The key findings for SB are summarized as follows:

Micro-CT imaging revealed significant enhancement in porosity, with SB exhibiting a notable 161% increase. This substantial porosity gain, driven by SB's inherent high permeability and initial porosity, facilitated extensive dissolution processes and effective mineral carbonation.

SEM analysis highlighted enhanced porosity resulting from olivine dissolution along crystallographic planes and microfractures, contributing to improved pore connectivity and overall reactivity.

XRD results identified the emergence of secondary mineral phases, reflecting active geochemical transformations and secondary crystallization.

Overall, the scoriaceous basalt demonstrated considerable potential for permanent mineral storage of CO_2, supported by its enhanced reactive surface area, interconnected porosity, and efficient carbonate precipitation dynamics, making it an exceptionally favorable medium for CO_2 mineralization processes.

Acknowledgements

The authors gratefully acknowledge the College of Petroleum Engineering & Geosciences providing the necessary facilities and resources for this research. We extend our appreciation to Mr. Bander Al-Otaibi and Mr. Ajibola Hakeem for XRD and SEM analysis. We acknowledge the support from KFUPM Hydrogen Consortium Grant H2FC2313, Usage of Alkali Basalt and Peridotite for CO_2 Capture, Utilization and Storage (ABPECUS).

References

Ahmed Ali, K., Ahmad, M.I., & Yusup, Y. (2020). Issues, Impacts, and Mitigations of Carbon Dioxide Emissions in the Building Sector. *Sustainability*, **12**(18), 7427. https://doi.org/10.3390/su12187427

Boruah, A. (2025). Climate Change and CO2 Geosequestration (pp. 1–17). https://doi.org/10.1007/978-3-031-81021-3_1

IEA. (2023). CO2 Emissions in 2022. IEA (2023), CO2 Emissions in 2022, IEA, Paris https://www.iea.org/reports/co2-emissions-in-2022, Licence: CC BY 4.0

Kelemen, P., Benson, S.M., Pilorgé, H., Psarras, P., & Wilcox, J. (2019). An Overview of the Status and Challenges of CO2 Storage in Minerals and Geological Formations. *Frontiers in Climate*, 1. https://doi.org/10.3389/fclim.2019.00009

Murtaza, M., Whattam, S., Fawad, M., Saraih, N.A., Kamal, M.S., Abu-Mahfouz, I.S., Hussain, S.M.S., & Kaminski, M.A. (2024, April). Assessing CO2 Mineralization and Sequestration Potential in Saudi Basaltic Rocks. https://doi.org/10.5194/egusphere-egu24-20955

Nunes, L.J.R. (2023). The Rising Threat of Atmospheric CO2: A Review on the Causes, Impacts, and Mitigation Strategies. *Environments*, **10**(4), 66. https://doi.org/10.3390/environments10040066

Olajire, A.A. (2013). A review of mineral carbonation technology in sequestration of CO2. *Journal of Petroleum Science and Engineering*, **109**, 364–392. https://doi.org/10.1016/j.petrol.2013.03.013

Raza, A., Glatz, G., Gholami, R., Mahmoud, M., & Alafnan, S. (2022). Carbon mineralization and geological storage of CO2 in basalt: Mechanisms and technical challenges. In *Earth-Science Reviews* (Vol. **229**). https://doi.org/10.1016/j.earscirev.2022.104036

Redlin, M., & Gries, T. (2021). Anthropogenic climate change: the impact of the global carbon budget. *Theoretical and Applied Climatology*, **146**(1-2), 713–721. https://doi.org/10.1007/s00704-021-03764-0

Whattam, S.A., Azizi, H., Iskandar, O., Osman, M., Heaton, D., Nouri, F., Lemdjou, Y.B., & Koppers, A. (2025). Asymmetric magma chamber beneath the Red Sea system controlled Cenozoic alkaline magmatism in western margin of the Arabian Shield. *Journal of the Geological Society*. https://doi.org/10.1144/jgs2024-280

Ye, H., Liu, Q., Bao, Q., Wang, Z., Xie, Y., Michelle, T., Zhao, W., & Xian, C. (2025). Review on in-situ CO2 mineralization sequestration: mechanistic understanding and research frontiers. *International Journal of Coal Science & Technology*, **12**(1), 15. https://doi.org/10.1007/s40789-025-00755-8